AI-Integrated Multi-Modal Academic Evaluation, Performance Intelligence, and Automated Viva Assessment System

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Project Proposal Report

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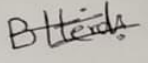
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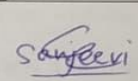
DECLARATION

Declaration

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Viva examinations are a common phenomenon used to assess the knowledge, communication skills, and confidence levels of the students. However, the conventional viva system mostly depends upon the judgment of the human evaluators, which may get affected by factors such as fatigue, cognitive overload, preconceived notions, and extensive review of the responses. To overcome these challenges, the research work proposes a system for the viva evaluation of the students, which makes use of the AI system for the analysis of the responses of the students. The system makes use of the audio and video responses of the Viva Examinations for the analysis of the confidence levels of the students, as well as the effectiveness of the responses provided by the students. The system makes use of deep learning, speech recognition, natural language processing, and computer vision for the analysis of the responses of the students. The facial expressions of the students are analyzed using the MediaPipe and DeepFace libraries, whereas the audio responses are converted into text using the OpenAI Whisper library. The audio features are used for the analysis of the emotions of the students, which are achieved using the Librosa and Parselmouth libraries. The second marking system is used to ensure the reliability of the results, as well as the academic integrity of the system. The system can also be used for online interviews, professional certifications, and skill assessments.

Keywords: artificial intelligence, viva evaluation, multimodal analysis, speech emotion recognition, computer vision, Video Analysis, Facial Expression Analysis.

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LIST OF ABBREVIATIONS

Table 1:Abbrevations

ML	Machine Learning
AI	Artificial Intelligence
DL	Deep Learning
NLP	Natural Language Processing
RF	Random Forest
XAI	Explainable Artificial Intelligence
DeepFace	Deep learning face recognition library
Wav2Vec2	Wave2Vec 2.0
FFmpeg	Fast Forward MPEG
OpenCV	Open-Source Computer Vision Library

1. INTRODUCTION

1.1 Background and Context

This research addresses the AI-Enhanced Online VIVA Assessment Solution as a significant aid to the current VIVA process. Different types of artificial intelligence technologies may be used individually to process the audio, video, and gestures of data that have been gathered during the VIVA assessment process. This research Aims to propose that AI technologies may be used to assess the student performance on traditional aspects such as emotional engagement, communication confidence, and so forth.

Additionally, the emotional condition of each student may be identified based on the analysis of their facial expressions, which may be happy, fearful, angry, and etc. emotion of the students based on their facial expressions, the analysis of the pitch, tone, and speech rates may be used as additional indicators of the emotional condition of the student with respect to their confidence levels in communicating their thoughts and/or feelings and gestures and moments. By utilizing natural language processing, the accuracy of the student's response could be captured.

This research is expected to assist the academic community by improving the process of assessing students through VIVA processes. AI has the potential to provide the academic community with many benefits through the use of AI technology for assessing students through VIVA processes.

Research Objective

1. To study the limitations of traditional practices of the viva examination, emphasizing the causes of subjective grading by the examiners.

2. To study various behavioral and communication skills which affect the confidence level of the students, such as confidence, facial expressions, tone of voice, and clarity of response in viva examinations.

3. To develop an artificial intelligence-based framework that can analyze the facial expressions, speech patterns, and responses of the students to assess their performance in viva examinations.

4. To study the efficacy of an artificial intelligence-based viva examination system in the context of contemporary education.

5.To Analyze student speech-to-text translation and analyze the tone of voice, clearance of the speech, and ability of expression speech thought the viva

6. To analyze student body gestures, body moment which are relevant to student facial and vocal output and measure

1.2 Literature Review

Automated Evaluation Technologies

Inventive advancements in the fields of artificial intelligence (AI), machine learning (ML), and computer vision (CV) have provided avenues for developing automated evaluation tools that replicate the human observation process and are able to process and evaluate audio, visual, and written information simultaneously. Research conducted in this field shows the analysis of audio, visual, and written data to provide a more comprehensive behavioral analysis than analyzing each separately. By processing multiple types of information simultaneously, these tools can identify cues of confidence, hesitation, emotional involvement, and comprehension.

Speech Emotion Recognition and Voice Analysis

Speech emotion recognition (SER) allows educators to objectively assess their student's mental state by examining several different characteristics of speech, including pitch, tone, rate of speech, and rhythm of pauses [1] [2] [3]. According to the research conducted by Sumayah and Pati (2025), the characteristics of speech are able to indicate levels of confidence along a grading scale. In addition to measuring levels of confidence, it has been found that characteristics of speech can more accurately reflect how well a student understands a topic compared to written information. For example, the authors found that differences in speech reflect doubt in students even when they have given a technically correct answer. To help improve the accuracy of student evaluation systems, modern-day assessment is performed using multimodal deep learning models, which evaluate both the characteristics of speech and written text.

A technology stack is crucial for the success of any implementation of proposed speech-based student evaluation systems. A technology implementation framework is composed of a series of components that use artificial intelligence (AI) libraries to operate the speech recognition part of the speech-based evaluation. The following are components in the technology implementation framework such as,

1) **Transcription**: The technological implementation framework will include use of the OpenAI Whisper library for transcription based on quality inputs from individuals' voices, to carry out the speech recognition portion of the evaluation function accurately and effectively

[4]. Prior tests showing the capability of this library in terms of accurately transcribing speech produced by individuals with various accents (including Sri Lankan accents), have all shown a high degree of accuracy.

2) **Feature Extraction:** The AI libraries, Librosa and Parselmouth, will work in conjunction to provide feature extraction of all audio-based responses to be used in determining the relationship between the student's confidence level in giving their responses during the evaluation process to their actual performance on specific tests. Both libraries are created using Python programming languages.

3) **Affect Modelling:** SpeechBrain and Wav2vec2 libraries were used to create an AI neural network model.

Facial Expression Recognition

Recognizing facial expressions is a critical aspect of understanding how people feel emotionally. Using computer vision technology, computers can now monitor people's facial movements automatically for purposes of recognizing the emotions they express. One of the types of deep learning technology that is being used for facial expression identification using computer vision techniques is: VGG16, ResNet50, and DenseNet [5]. This outlines a sampling of tools that have been employed to detect and reject facial expressions for use with recognized primary emotions: happiness, fear, anger, surprise, sadness, and disgust.

Current computer-based applications often employ a combination of convolutional neural networks (CNN)–long short-term memories (LSTM) systems in order to record and analyze not only an individual’s current facial expressions at any given point in time, but also their prior facial expressions over a period of time. This allows for the analysis of the dynamic changes in a person’s face throughout their entire lifespan, but other studies supported by the research have documented an ability of deep learning tools to achieve in excess of 90 percent accuracy for identifying primary emotional expressions from individuals. However, a significant gap between people’s emotional states and their level of academic confidence exists towards establishing and maintaining the appropriate relationship between their emotions and their educational outcomes.

Natural Language Processing for Content Evaluation [6]**.**

Although behavioral responses provide feedback on student performance, what actually counts for assessing students is the content of their written response. Therefore, to assess the meaning

of a student's response in terms of its logical structure, Natural Language Processing (NLP) will be applied. In the past, traditional automated assessment systems relied heavily on the use of keywords to evaluate answers. These systems did not effectively evaluate the students' answers. However, with newer language models such as BERT and RoBERTa, machine learning tools can effectively understand human language through the relationships between words in context.

The assessment of student answers can rely on the student's response being effectively evaluated by meaning, logic, and content, rather than by the mere presence of keywords. Research has identified an automated way of evaluating descriptive answers using techniques such as Word2Vec, Word Mover Distance, and Cosine Similarity to provide students with a comparison for their answer [7]. The accuracy of their evaluation method is rated at 88 percent in comparison to a list of validated reference answers. TF-IDF is another common example where an answer is converted into numerical vectors.

In earlier iterations of automated assessment, keyword matching was the principal method used to evaluate responses from students. The drawback of that method was the inability to provide a good indication of a student's correct answer because the method relied solely on keyword presence. What has changed since then is the development of language models like BERT and RoBERTa, which allow for increased accuracy when it comes to the interpretation of text using machine learning systems with and between understanding of human language based on context. Therefore, responses are evaluated for their semantic meaning, reasoning, and conceptual correctness rather than about whether or not they contain certain words.

Benchmarks and Datasets for Viva Assessment

To create and assess AI systems used for emotion recognition, speech analysis, and emotion evaluation, a variety of sets of reliable and helpful datasets are needed to accurately represent each of the key components involved in developing these types of systems. These databases include data from other studies that were performed on emotion recognition, speech analysis, and emotion evaluation; however, the datasets have been created specifically for measuring emotional expressions via facial recognition [8]. For instance, MediaPipe is one such system that was created specifically for evaluating facial movements from a database of 3d facial landmarks. Similarly, DeepFace was created to evaluate emotional state(s) using a pre-trained model. Finally, the PyFeat library has been created specifically for evaluating facial features as defined by the facial action coding system (FACS) [9].For instance, MediaPipe is one such

system that was created specifically for evaluating facial movements from a database of 3d facial landmarks. Similarly, DeepFace was created to evaluate emotional state(s) using a pre-trained model. Finally, the PyFeat library has been created specifically for evaluating facial features as defined by the facial action coding system (FACS) [9].

Facial Expression Datasets

The FER2013 dataset is one of the most frequently used datasets to execute facial expressions. FER2013 consists of 35,887 greyscale images, each of size 48 × 48 pixels, depicting facial expressions. 7 classes of emotion represented in FER2013 represent the look of a person and the emotions shown from the look. The FER2013 dataset was originally created for a competition hosted on the Kaggle platform.

AffectNet is another commonly used dataset to perform facial expression recognition. Over 1 million images of people exhibiting different facial expressions captured via the internet have been stored here. Each facial expression is classified into the respective categories of emotion exhibited by the subject. Another major dataset for performing facial expression identification is CK+ (Cohn-Kanade Plus). The CK+ dataset consists of 600 video clips that depict a person's facial expressions changing from neutral to peak expressions. The video clips are used to classify the emotions of the subjects using the Facial Action Coding System (FACS).

Multimodal Emotion Datasets

Multi-modal datasets offer researchers the capability of examining the emotional expressions of participants through the use of multi-modal recordings that contain both audio and video content. The two most widely used datasets have been the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS), consisting of 4904 videos of actors portraying many different emotions including calmness, happiness, sadness, anger, fearfulness, surprise, and disgust and the MELD (Multimodal Emotion Recognition Dataset) - a multi-modal dataset that provides both audio and video representations of the emotional state of participants engaged in multi-party conversations. The RAVDESS dataset is largely used for the purposes of developing and testing emotion recognition models due to its inclusion of both acoustic and visual elements produced by the participants. Abdullah et al. utilized the visual elements from the RAVDESS dataset for the development of their CNN-LSTM models, which produced about 61% accuracy in predicting the emotional states of the participants. The MELD dataset is an expanded version of the Emotion Lines dataset that incorporates audio, video and text elements

to support the creation of AI-based models designed to analyze emotional expression and detect behavioral indicators throughout evaluation process.

Viva-Specific Datasets

Even though these types of datasets can be useful when training a model, for research conducted mainly on the academic viva exam, the datasets used should be representative of the actual questions and answers given during an academic viva exam between the student and the examiner. One type of dataset is the Custom Viva Response (CVR). The CVR Dataset consists of 685 Q&A pairs. Each Q&A pair contains the question the examiner asked, the student's response, and a degree of confidence as rated by the OCR of the speech transcript.

The confidence values are determined by speech attributes that reflect how the student responded given different types of disfluencies (e.g., filler words, fragmented sentences, hesitation markers such as "I think" or "maybe"). Another type of academic dialogue dataset is the Amrita viva corpus, which is a 348-minute-long collection of audio recorded viva examinations of 40 students. The audio corpus consists of 685 answers provided by students to a variety of questions posed by the examiners, together with markup for the students' confidence shown in answering the questions. This dataset of academic dialogue recorded spontaneity in the way that the students displayed their comprehension of difficult academic content.

Question Answering and Educational Datasets

In addition to emotion recognition datasets, question answering datasets are also useful to evaluate conceptual understanding. The Stanford Question Answering Dataset (SQuAD) is a major source for developing reading comprehension and question-answering models. The dataset has many thousands of questions created using Wikipedia and any associated answers from the relevant section of Wikipedia where that question can be found that the fine-tuning of a BERT-based model on this dataset resulted in a correct-response rate of 92% [10].

The MSVD-QA dataset was created to provide a way of expanding the use of question-answering methodologies into the video domain. The MSRVTT-QA dataset is likewise a way to expand the use of question answering the video domain. These datasets contain questions about the same video files as the Microsoft Research Video Description dataset. These datasets will enable the development of question-answering models that can deal with both text and video input.

AI-Based Interview and Viva Evaluation Systems

The main focus of utilizing these systems is to improve consistency and the reliability of evaluating and analyzing the candidate response automatically to increase their ability to communicate effectively, candidates can use many different kinds of AI-based platforms that have recently become available for the conduct of mock interviews. Typically, an AI platform employs speech recognition and natural language processing methods to evaluate users' responses to interview questions, much like a real interviewer would [11].

In addition to facial emotion recognition and speech emotion recognition included semantic content analysis into the interviewee's facial expression recognition system (FaceExpressionNet), voice recognition system (Mel-Frequency Cepstral Coefficients), and text analysis system (transformers). Demonstrated accuracies of 89.8% for facial emotion recognition, 87.6% for voice emotion recognition, 92.3% for semantic matching.

There is a system developed by an AI-based interview evaluation system but used a different approach [12]. They employed natural language processing and machine learning techniques to evaluate candidates' responses to questions. Used speech recognition to convert oral responses into text, the **Levenshtein** distance method to match keywords between candidate responses and the interviewer's specified answer, and logistic regression to assess personality characteristics of candidates in order to predict how they would respond to the same question.

Although all of these studies demonstrate the potential of using AI-based interview evaluation systems to evaluate candidates' responses, these types of systems are primarily oriented towards human resource management applications, such as recruitment interviews, and not academic application types of academic exam. As a result, existing systems primarily focus on evaluating candidates' responses based on their communication skills rather than their conceptual understanding.

Assessment of Student Confidence

Various studies have been conducted on the automatic estimation of student confidence levels while giving their oral responses. In one research paper presented a machine learning-based framework that uses the embedding techniques TF-IDF, BERT, RoBERTa, and GloVe to transform the textual representations into numerical representations. Further, the features were used to train classification models Multilayer Perceptron, AdaBoost, and Logistic Regression to classify the student's response into respective confidence levels. The authors have shown

promising results from the experiments conducted, which prove that the multilayer perceptron classifier with GloVe features is the best combination to accurately predict the student's confidence level with an accuracy of 76 percent. Another research paper developed a confidence assessment model based on the acoustic features of the student's speech. By using the MFCC, DCT, and LPC features under the multi-feature fusion framework, the authors were able to attain a classification accuracy of 87.54 percent [13].

Optimization and Technical Architectures

To implement automated viva evaluation techniques, an integrated computer system is needed which can process audio, video, and textual information. Currently, many computer systems have been created using a combination of deep learning algorithms: convolutional neural networks (CNNs) used to process video images (extracting spatial features), while long short-term memory networks (lstms) process audio or video (extracting temporal features). Since video processing is computationally complex, several optimization techniques have been introduced to increase the accuracy of video analysis. An example of optimization technique for video analysis is key frame extraction which involves extracting the video sequence of the significant frames. Network pruning is another method of optimizing video analysis using deep learning algorithms. Principal Component Analysis (pca) is also used as a video analysis technique for the reduction of dimensions. OpenCV and ffmpeg are two common video analysis packages.

Explainable AI and Ethical Considerations

Main problem associated with using AI to determine a student's grade is that there isn't much transparency in the automated decision-making process. Therefore, it is vital for both teachers and students to know how an AI model came up with a particular grade. Explainable AI (or xAI) attempts to address the problem of transparency by giving some explanation for the predictions made by AI models. The LIME and SHAP techniques provide information about which input features contributed to a model's prediction. To establish a visual explanation of the automated decision-making process, a framework based on using CLIP-based video question answering and YOLOv8 for object detection [14]. In this framework, the explanation for an evaluation will be obtained from the video frames that were used in the evaluation.

Research Opportunity

The literature review contained in this chapter pointed out that there exist great possibilities with the use of AI for enhancing the automated assessment process; however, thus far, there exists no overall framework developed specifically to evaluate academic Vivas using multimodal means.

Future studies can fill in the gap left by this study and create a multimodal, holistic evaluation framework that incorporates technical correctness, behavioral confidence and emotional engagement through speech, facial expression and natural language content. This type of system could be developed using multimodal data fusion and explainable artificial intelligence to create an automated assessment framework that assists both in evaluating student performance and in providing students with meaningful, timely and uniform feedback.

1.3 Research Gap

The pivotal intention of this research is to mitigate the fatigue of teachers after doing viva long time a day. It reduces the teacher's correct mindset also because of fatigue [15] [16]. Significant research has been done to explore the application of emotion recognition and audio analysis to evaluate communication skills, interview skills, and emotionality. However, most artificial intelligence-based evaluation tools are based on a specific criterion or modality either the analysis of verbal responses using speech-to-text or Natural Language Processing (NLP), or the analysis of non-verbal responses such as facial expressions or micro-expressions to detect mood [17]. Though such tools can be used to evaluate grammatical skills, vocal skills, or emotionality, they are not based on the evaluation of both audio and video data. The evidence suggests that the evaluation of multiple modalities can improve the reliability of emotionality.

For instance, recent AI-based mock interview tools demonstrate the capabilities for facial expression analysis and automated feedback. Nevertheless, these tools appear to be more aligned towards a practice environment rather than formal academic assessment modes such as viva voice. The viva assessment process requires simultaneous evaluation of technical knowledge, logical thinking, communication skills, and confidence. Furthermore, the majority of the available tools do not offer an explainable outcome, thereby failing to offer a deeper understanding to the user about the factors affecting the assessment process.

It can be stated that a notable gap exists in the development of a comprehensive and multimodal AI-based viva assessment system. The study aims to address this gap by proposing a comprehensive AI-based system that can simultaneously process both audio

and video streams for confidence and technical accuracy while incorporating an explainable AI model for generating detailed and comprehensible performance reports.

Table 2: Research Gap

	My research	Research [1]	Research [3]	Research [7]	Research [9]	Research [15]
Academic engagement	*yes*	*yes*	*yes*	*No*	*No*	*No*
Explainable output	*yes*	*No*	*No*	*Yes*	*Yes*	*No*
Audio analyzing	*yes*	*yes*	*yes*	*yes*	*No*	*Yes*
Video analyzing	*yes*	*No*	*No*	*yes*	*yes*	*Yes*
Evaluate Grading	*yes*	*yes*	*Yes*	*Yes*	*No*	*Yes*

2. OBJECTIVE

2.1 Main Objective

The main goal and objective of this study is to identify how an AI-generated Viva evaluation system can be utilized by the examiners in an efficient manner in relation to the manual process while at the same time improving the morale of the students and ensuring effective and efficient grading.

2.2 Specific Objective

- To design a computer vision-based module that can identify students' emotional states, facial expressions, and eye gaze patterns during viva voce sessions.
- To design a vocal analysis module that can identify different vocal characteristics such as pitch, speech rate, tone, pauses, and energy levels in students' answers and estimate students' confidence and fluency.
- To design a module that can convert students' answers to text and evaluate the relevance, coherence, and technical accuracy of students' answers using natural language processing.
- To combine the acoustic prosodic with the semantic of the transcribed answers to evaluate the confidence levels of the students and the clarity of their articulation.
- To design an evaluation engine using the principles of natural language processing to identify conceptual misconceptions and perform semantic similarity assessments relative to the reference material provided to grade subjective answers.
- To design an evaluation engine using the principles of explainable artificial intelligence to provide transparent criteria to the assigned grades.

Alignment Justification:

- **Measurability**: Each objective targets a specific technical output (e.g., "fusion of features," "CNN-LSTM implementation," "XAI integration") which can be tested for accuracy and performance.
- **Problem Alignment**: Addresses the issues of "examiner fatigue," "subjectivity," and "scalability" identified in your research problem

3. METHODOLOGY

Viva analysis system involved in AI based utilized the Design Science Research (DSR) methodology alone with experimental validation. The main focus is to address real-world problems in academic assessment. In this study the process will be followed commencing on the problem identification, solution objectives, design and development of the system, demonstration of the system, Evaluation and the communication by using this DRS system. To analyze the captured frames of a video processing pipeline can be used. For the start, face detection and Landmark extraction will ease the process; the frames will be extracted at 1fps ensuring the efficiency of the prices. Technologies like OpenCV and FFmpeg can be utilized for that. Moreover, face detection technologies like mediaPipe or Fscemesh can be used which are captured in 3D facial landmarks detection compared to the captures in normal cameras. So, technology like deep faces can be used which enables you to detect 7 cases of emotions.

Audio processing pipelines are being used to process the audio of the students in viva sessions. For the process or audio, tools like FFmpeg, Librosa can be used to capture it accurately. And to capture the pitch and analysis it, we can use Parselmouth which can measure energy, speak rate and pause patterns precisely. Moreover, to analyze the speech emotion classification from vocal characteristics tools like SpeechBrain, Wav2Vec2 can be used. Those tools are capable of frequently extracting and filtering the words like (“uh”, “um”, “hmm”). And the option for filtering the word patterns is also vital for this system. For example, “maybe”, “perhaps”, “I think”, “sort of”. Audio filter detection also can match the identified patterns.

In this process, by using the Speech-to-Text and NLP Pipeline component, the confidential scores and performance grades will be provided for each student by using OpenAI Whisper with technical term accuracy. We can use semantic embeddings like BERT and Roberta to semantic Understandings. Hedge word detection finds pattern mismatch by detecting uncertain phrases like ("maybe","i think", "perhaps"). For multimodal fusion and strategy, the utilization of Random Forest as Beta models and AdaBoost as Meta-Model will be an advantage.

Finally, to execute the validation of Strategy and performance metrics by utilizing the datasets which can be collected primarily from Student Viva evaluation Recording and then can be evaluated by the ai base grading. We can use RAVDESS from Ryerson University

which has around 4000+ videos for multimodal emotion recognition. In Kaggle FER2013, it consists of a facial expression model dataset.

Table 3:Technologies

Model	Usage
Video Processing Pipeline	
OpenCV	Video capture, frame extraction, and basic image processing operations.
FFmpeg	Handles video format conversion and ensures consistent resolution
MediaPipe Face Mesh	real-time detection of 3D facial landmarks with high accuracy
DeepFace	facial expression recognition (classifies seven emotions)
CNN-LSTM Architecture	extracts spatial features from individual frame
Audio Processing Pipeline	
Librosa and Parselmouth	pitch tracking and formant analysis
SpeechBrain, Wav2Vec2	Speech emotion recognition
OpenAI Whisper	selected for speech-to-text transcription
Overall implementation LLM	
RoBERTa	Optimized BERT variant
AdaBoost, KNN, Random Forest	Combines base predictions for final score
XAI	Explainable AI
DATASET	
Kaggle	
RAVDESS	
Amrita Viva Corpus	

4. HIGH-LEVEL SYSTEM ARCHITECTURE

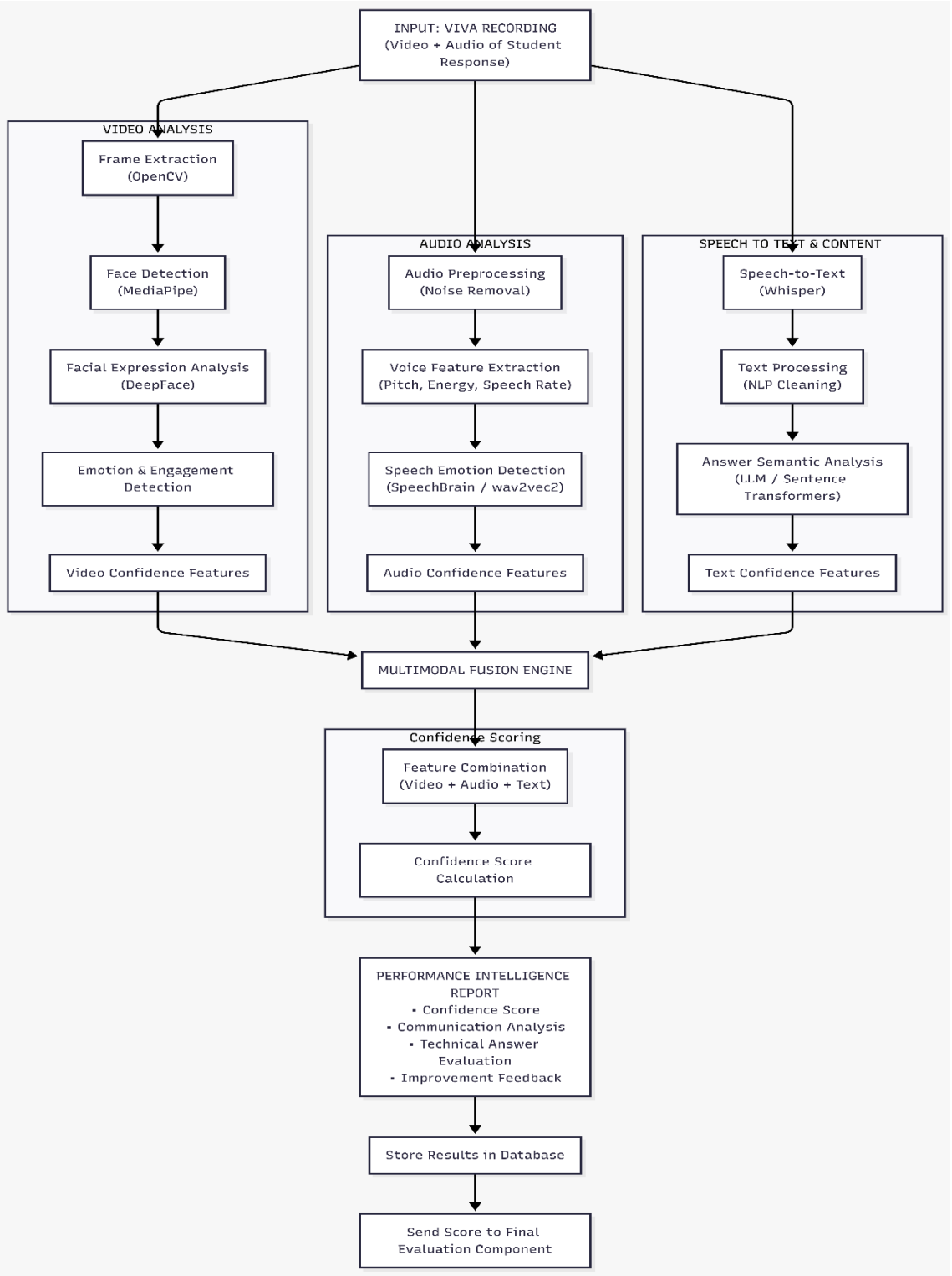


Figure 1: highlevel architecture

5. USER REQUIREMENTS

- Lectures
- Examiners
- Students

5.1 Functional Requirements

- System must be able to accept Videos and audios data as inputs.
- System must be able to detect Video students face in a video.
- System must be able to detect internal damages using sensor inputs.
- System must be able to assess the severity and the type of damage.
- System must be able to decide whether the part should be replaced or repaired.
- System must be able to calculate the claim amount accordingly.
- System must be able to store history in cloud storage for later use.

5.2. Non-Functional Requirements

• **User - Friendliness:** - The system should be simple and user friendly, due to the stressful conditions the application is used in.

• **High Availability:** - The system should at least 99.8% be available given the criticalness of the application.

Security: Security of the data at rest and in transit must be ensured by encryption and industry standards procedures.

• **Scalability**: - The system should be horizontally scalable to facilitate the growing number of users.

Real time: The claim estimates must be given in real time, with no delay.

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

The main intended beneficiaries of the proposed AI-Integrated Multi-Modal Viva Evaluation System are educational institutions, online learning platforms, and professional certification bodies that use oral assessments and evaluative interviews as part of their processes.

Primary customers:

- Private and Government Universities
- Higher educational institutes conduct presentations and via sessions.
- Online education platforms offering remote learning and assessments.
- Professional certification organizations which conduct oral examinations.
- Corporate training institutes which evaluate candidate's communication skills.

Customer Persona:

University Academic Coordinator

Role: Lecturer / Examiner

- Responsibilities: Managing student assessments and viva examinations
- Needs: Efficient, fair, and scalable evaluation methods for large student groups
- Pain Points: Time-consuming manual viva evaluation, inconsistent grading, lack of detailed feedback

6.1 Value Proposition

The proposed system provides an AI-based solution to the problem of viva voce examination through the analysis of both verbal and non-verbal cues. The value proposition of the proposed system can be listed as follows:

• Objective evaluation: This helps to remove the bias or subjectivity involved in the examination process.

• Automated performance analysis: This helps to analyze the confidence levels, emotional involvement, and communication skills of the candidates.

• Scalability: This solution can be implemented on a large scale to perform viva voce examinations in universities.

• Detailed feedback: This solution provides performance intelligence to the candidates as well as instructors.

• Time efficiency: This solution can save the time of the examiners involved in the viva voce examination process.

The proposed solution provides a more comprehensive evaluation of the candidates through the analysis of speech, facial emotion recognition, and semantic evaluation.

6.2 Revenue Model

There are a number of ways to do so. However, all of them maintain the same ideas and details.

- SaaS model: Schools and other educational institutions can subscribe to the system by paying a fee on a monthly or yearly basis.

- Pay-per-assessment model: The fee can be charged according to the number of viva sessions done by the system.

- Licensing model: Universities can obtain a licensed version of the system.

- API integration services: Online education platforms can be integrated into the evaluation system using API.

6.3 Cost Estimation

The main costs associated with developing and maintaining the system include:

Development Costs

- AI model development and training
- Software development (frontend and backend)
- Integration of computer vision and speech processing modules

Operational Costs

- Cloud computing infrastructure for video and audio processing
- Data storage and security
- AI model maintenance and updates

Estimated Monthly Costs

- Cloud computing and processing: $200 – $300
- Data storage and server maintenance: $100 – $200
- Software updates and maintenance: $50 – $150

Total estimated operational cost: $500 – 600 per month depending on system usage.

6.4 Pricing Strategy

The pricing strategy is designed to be affordable for educational institutions while maintaining sustainable revenue.

Example pricing structure:

Basic Plan

- For small and medium institutions
- Up to 100 Viva evaluations per month
- Approximate price: $50 – $80 per month Professional Plan
- For universities and large-scale institutions
- Up to 1000 Viva evaluations per month
- Approximate price: $100 – $200 per month

Enterprise Plan

- Unlimited evaluations with custom integrations
- Pricing based on institutional requirements

A freemium model may also be introduced for universities to test the platform before purchasing a full license.

6.5 Competitive Advantage

The proposed system offers several advantages compared to the existing interview analysis platforms.

• Multimodal evaluation: combines facial expressions, vocal expressions, and the response itself.

• Academic-focused design: Designed for viva examinations rather than interviews.

• Automated confidence scoring: Measures student engagement and emotional behavior during presentations/viva.

• Performance intelligence reports: Provides detailed analytical insights rather than simple scores.

• Teacher-assisted verification - Allows lecturers to verify the scores generated by the AI system. All these features make it a unique tool for automated academic evaluation.

These features create a unique solution for automated academic evaluation systems.

6.6 Intellectual Property Considerations

The system may involve several intellectual property components.

• **Software copyright** - The software shall be subject to applicable copyright laws.

• **AI model ownership** - Custom-trained machine learning models used to perform confidence assessment and multimodal analysis can be considered a proprietary technology.

• **Patent potential** - The integration technique to combine facial emotion recognition, voice prosody analysis, and semantic answer evaluation to grade viva can be considered a candidate to be patented.

• **Data privacy & compliance** - Since the system handles video and audio content for the students, it is important to comply with data privacy laws to safely handle the information of the users.

7. BUDGET AND JUSTIFICATION

Table 4: budget

Item	**Justification**	**Amount**
Audio processing Tool	training deep learning models	12000
Video processing model	Used for video processing, AI model deployment, cloud storage, and scalable computing resources during testing.	14000
Internet bill	1500 * 8	12000
Deployment and hosting	Cloud platform for hosting	12000
Contingency Cost	For 8 months	1000
Total		60000

8. WORK BREAKDOWN STRUCTURE (WBS)

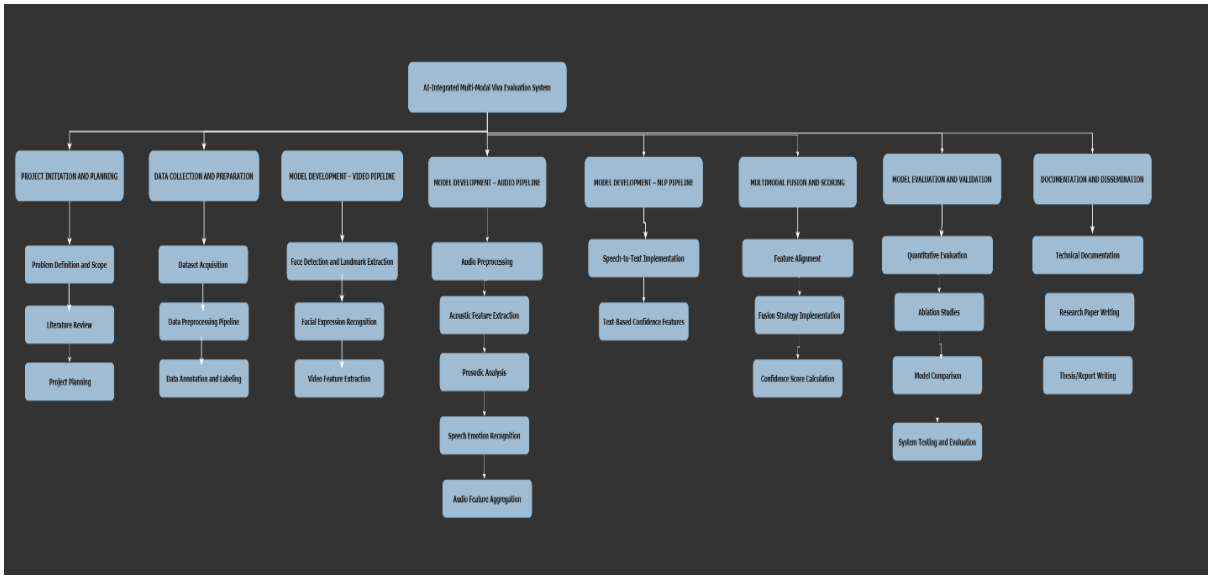


Figure 2: work breakdown structure

9. GANTT CHART

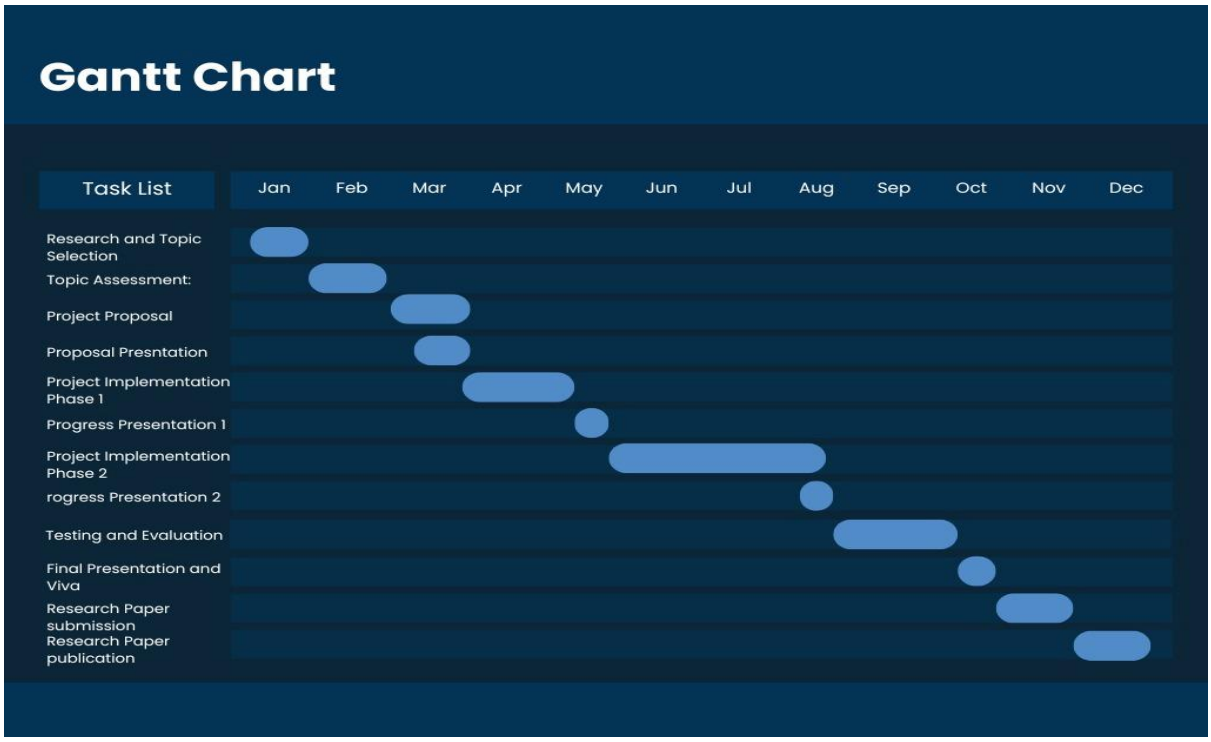


Figure 3: grant chart

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